

(1)

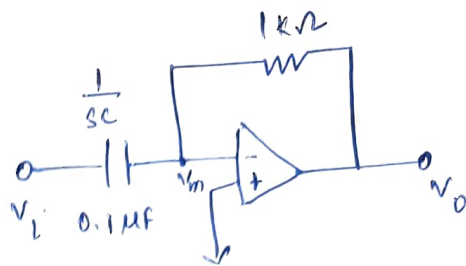
$V_m = 0$ (virtual ground)

Writing nodal analysis at V_m

$$sC(V_m - V_i) + \frac{V_m - V_o}{1000} = 0$$

on solving, we get

$$V_o = -0.8 \pi \cos 4000 \pi t$$



option:- D

(2) Diode is ideal

Let $V_i = 5V$

↳ assume diode is off

so, $V = 1V$ ↳ assumption is correct

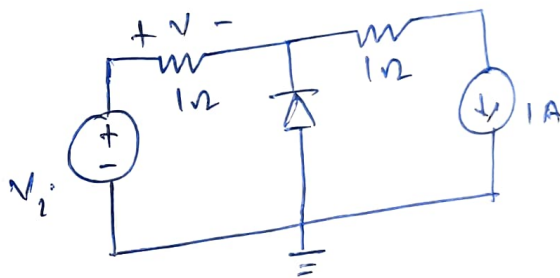
so, option (b) & (c) have been eliminated.

Let $V_i = -5V$ ↳ assume diode is on

$V = -5V$ ↳ assumption is correct.

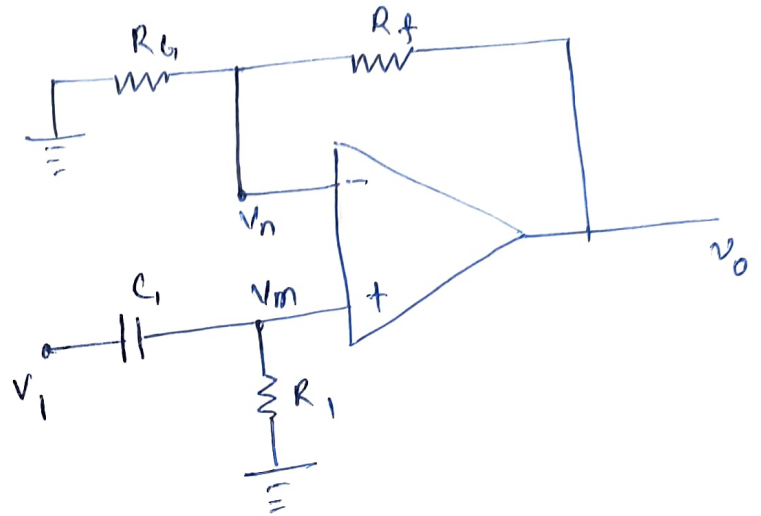
so, option (d) eliminated.

so, finally, option A



- ③ (E) (input resistance of opamp is ∞
+ve feedback configuration in opamp
operation is typically unstable).

④.



$$V_m = \frac{V_i R_1 s C}{1 + R_1 s C} \quad (\text{voltage divider rule})$$

$$V_m = V_n \quad (\text{virtual ground concept})$$

Nodal analysis at V_n

$$V_o = \frac{V_i s R C (R_G + R_f)}{R_G (1 + s R C)}$$

$$V_o = \frac{V_i R C (R_G + R_f)}{\left(\frac{R_G}{s} + R_G R C \right)}$$

Put $s = 0$ (Low Frequency) $\Rightarrow V_o = 0$

Put $s = \infty$ (High Frequency) $\Rightarrow V_o = \text{finite value}$

so, High Pass Filter

option:- B

⑤ $V_z = 0.7V$ (Given)

Assume D_1 ON & D_2 OFF

KVL (Left side)

$$15 = 5000i_1 + 0.7 + 10000i_1$$

$$i_1 = 0.953 \text{ mA}$$

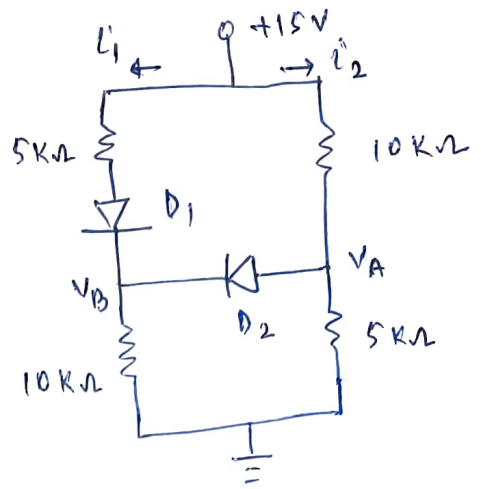
KVL (right side)

$$15 = 15000i_2$$

$$i_2 = 1 \text{ mA}$$

Now, $V_A = 5V$, $V_B = 9.53$ & assumption is correct.

so, option (c)



⑥

$$V_m = \frac{V_s \times 20}{40} = \frac{V_s}{2} \quad [\text{voltage divider}]$$

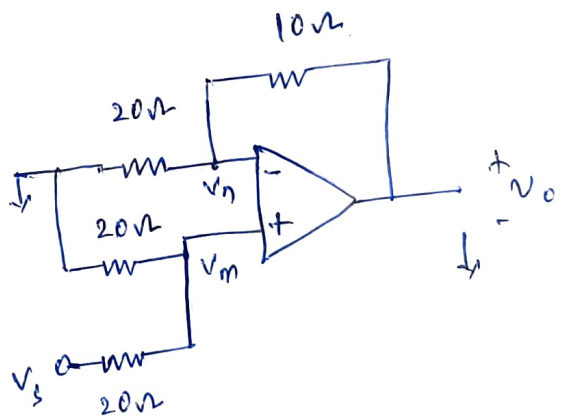
$V_n = V_m$ (virtual ground)

writing nodal analysis eqⁿ at V_n

$$\frac{V_n}{20} + \frac{V_n - V_o}{10} = 0$$

$$V_o = 0.75 V_s$$

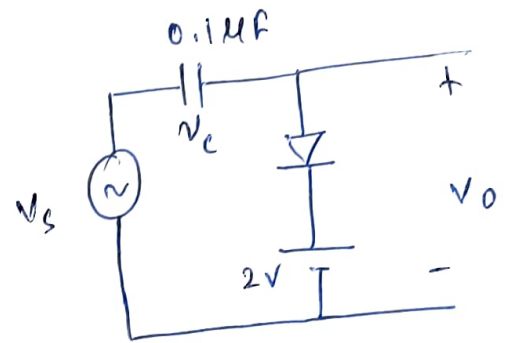
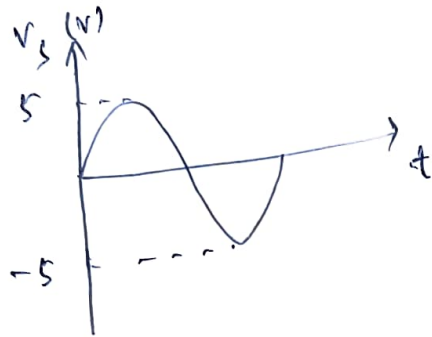
option A



7. (B)

8. $V_f = 0$ (ideal diode)

$$V_s = 5 \sin \omega t$$



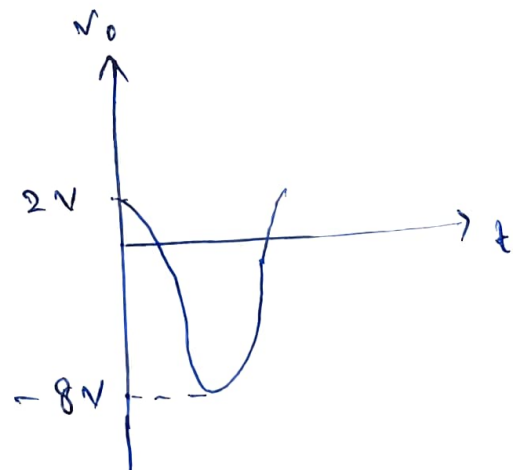
So, when the anode voltage of diode $> 2V$

Diode will be ON & $V_o = 2V$

As, this is a clamper circuit, so O/P peak to peak value will be $10V$ (same as i/p P-P)

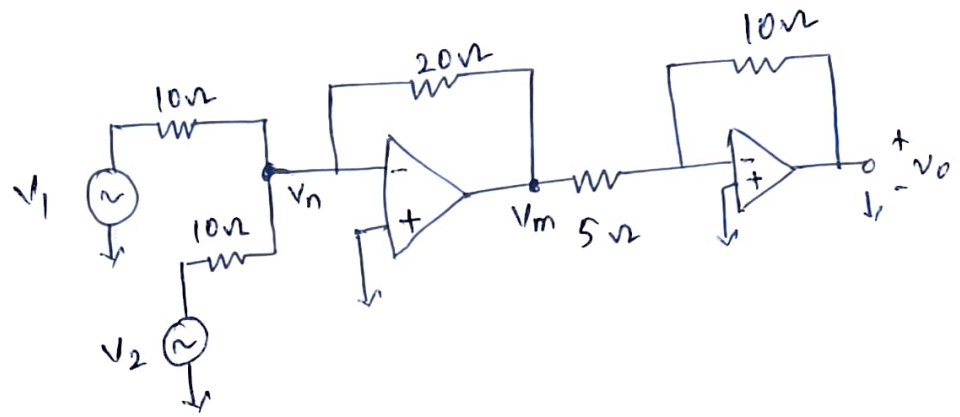
When the diode is off

then $V_o = V_s - V_c$



option :- (B)

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Given $V_1 = 2V$ & $V_2 = 3V$

Writing KVL from V_2 to V_1

$$V_2 = 10i + 10i + V_1$$

$$i = \frac{1}{20} A$$

$$V_n = V_2 - 10 \times \frac{1}{20} = 2.5 V$$

$$V_m = -\frac{R_f}{R_i} (V_n) = -\frac{20}{5} [10/110] \times 2.5 = -10 V$$

$$V_0 = -\frac{10}{5} \times V_m = 20 V$$

Ans.

(10)

Diode is ideal

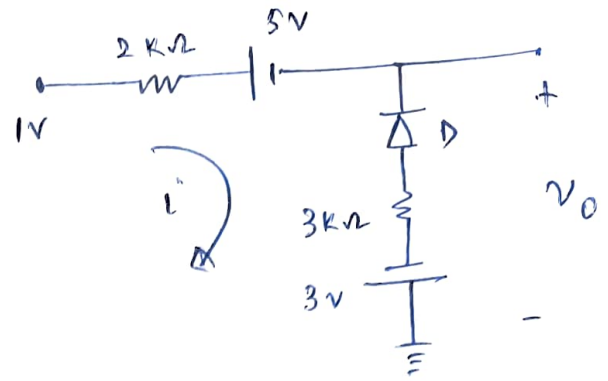
Assume diode is ON

KVL:- $1 = 5000i + 2$

$$i = -\frac{1}{5000} \text{ A}$$

Yes, assumption is right (current is flowing from anode to cathode)

so,
$$V_o = -\frac{3000}{5000} - 3 = -3.6 \text{ V}$$

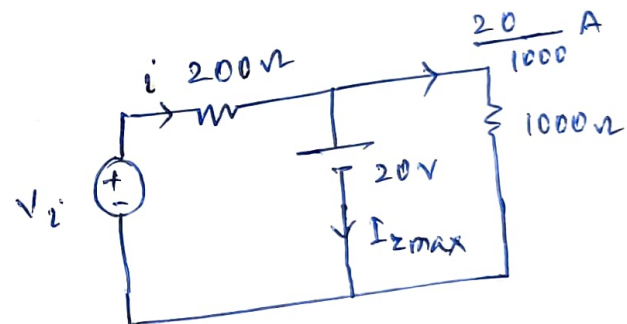


(11) Given, $V_Z = 20 \text{ V}$
 $I_{Z\max} = 60 \text{ mA}$

$$i = \left(I_{Z\max} + \frac{20}{1000} \right) \text{ A}$$

$$i = \frac{80}{1000} \text{ A}$$

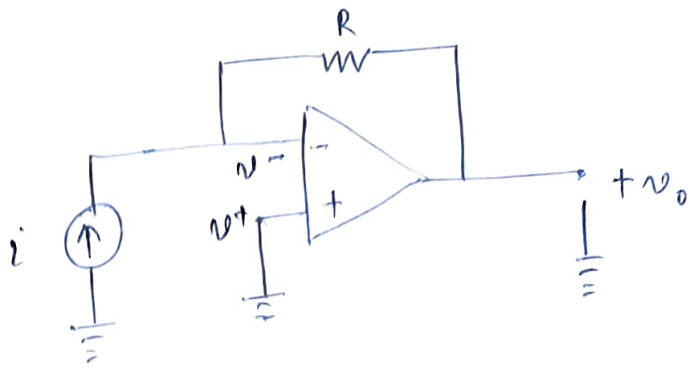
$$V_i = 200 \left(\frac{80}{1000} \right) + 20 = 36 \text{ V}$$



(12)

Given $i = 2A$

$R = 4000\Omega$



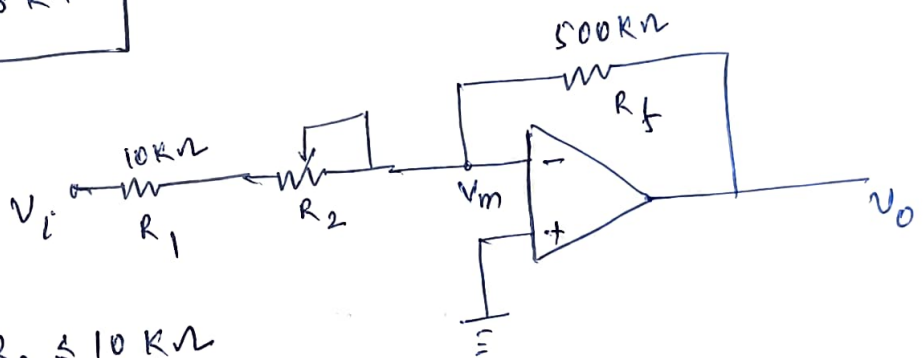
$v^+ = v^- = 0V$ (virtual ground)

Writing nodal eqⁿ at v^-

$$-i + \frac{v^- - v_o}{R} = 0$$

$$v_o = -8kV$$

(13)



Given $0 \leq R_2 \leq 10k\Omega$

$v_m = 0$ (virtual ground concept)

Writing nodal eqⁿ at v_m

$$\frac{v_m - v_i}{R_1 + R_2} + \frac{v_m - v_o}{500k\Omega} = 0$$

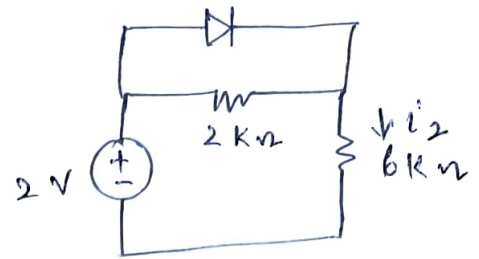
on solving

$$\left. \frac{v_o}{v_i} \right|_{\max} = -25 \quad \& \quad \left. \frac{v_o}{v_i} \right|_{\min} = -50$$

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$$V_T = 0.7 \text{ V (given)}$$

Assume diode is off



KVL

$$2 = 8000 i_2$$

$$i_2 = 0.25 \text{ mA}$$

& hence diode is off

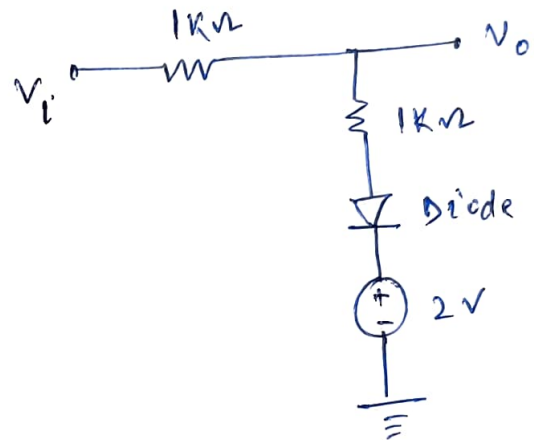
15

$$V_T = 0.7 \text{ V (given)}$$

$$V_i = 5 \sin \omega t \text{ V (given)}$$

$$\text{max} = 5 \text{ V}$$

$$\text{min} = -5 \text{ V}$$



Taking $V_i = 5 \text{ V (max)}$

Diode ON (Let)

KVL :-

$$5 = 2000 i + 0.7 + 2$$

$$i = \frac{2.3}{2000} \text{ A}$$

& diode is really ON

$$V_o = 3.85 \text{ V}$$

Taking $V_i = -5 \text{ V (min)}$

Diode off

$$V_o = V_i = -5 \text{ V}$$